SET-C

GLA UNIVERSITY,MATHURA

COURSE:B.TECH YEAR:II SEM:III Sec:A,B,C,D Branch:CS

I Mid Term Sub: Object Oriented Programming(ITE 211) Time:90 Minutes

Session:2012-13

Section A

Objective Type Questions **(8x0.5=4 Marks)**

1.Which of the following diagram is basically a flow chart to represent the flow from one activity to another activity.

- Collaboration diagram
- State Machine diagram
- Sequence diagram
- Activity Diagram

2. In UML, a state is graphically rendered as

- Rectangle
- Oval
- Lozenge shape
- Square

3. Multiplicity ``1..*'' represents

- Zero or more
- Many
- One or more
- Exactly one

4. In class diagram attributes are shown in

- First compartment
- Third compartment
- Second compartment
- Fourth compartment

5. _____ relationship is used to model 'Inheritance' property of Object Oriented systems

- Dependency
- Generalization
- Association
- Realization

6.Which features of the OOP showing the essentials details and hiding the non essential details.

- Encapsulation
- Abstraction
- Polymorphism
- Inheritance

7. Which of the following is used to model the life time of an object

- Use Case
- State Diagram
- Class
- Interface

8. UML is a

- Programming language
- Scripting Language
- Modeling Language
- Natural Language

Section B

Short Answer Type Questions(Attempt any two) **(2x3=6 Marks)**

Q1. With the help of proper illustrations differentiate between the concepts of ***association*** and ***aggregation***.

Q2. Draw the state machine diagram for money withdrawal from ATM.

Q3.What is Modeling ? Explain the types of models used in UML.

Section C

Long Answer Type Questions(Attempt any two) **(2x5=10 Marks)**

Q1. Discuss the features of the Object Oriented Programming?. What is Object Oriented Language? Name any two such languages.

Q2. Draw a class diagram for a information modeling system for a school.

- School has one or more Departments.
- Department offers one or more Subjects.
- A particular subject will be offered by only one department.
- Department has instructors and instructors can work for one or more departments.
- Student can enroll in upto 5 subjects in a School.
- Instructors can teach upto 3 subjects.
- The same subject can be taught by different instructors.
- Students can be enrolled in more than one school.

Q3 .WAP to create a class named as Bank, which should hold three variables Deposit, Withdraw and Balance; all of integer type. Create another class named as Operation, to accept the amount to be withdrawn by the user. If the amount to be withdrawn by the user is greater than the balance amount then a message "Invalid Transaction" should be printed on the screen.

GLA University, Mathura
I-Mid Term Examination, 2012-13

SET-C

Course: - B.Tech (CS) **II-Year, III-Sem.**

Subject: - Computer Organization (CSE-212) **Uni. Roll No:-**

Time:-90 Minutes **Total Marks:-20**

Notes:-

1) *Answer all questions; Section A is **compulsory**, **any two** from Section B and **Any two** from Section C.*
2) *All parts of a question (a, b, etc.) should be answered at one place.*
3) *Answer should be brief and to-the-point and be supplemented with neat sketches.*
4) *Any missing or wrong data may be assumed suitably giving proper justification.*
5) *Figures on the right-hand side margin indicate full marks.*

SECTION-A

Objective Questions (Write the correct answer) : **(8*0.5 = 4)**

I. N- bits can store in
- N+2 Flip-flop
- N Flip-Flop
- N-1 Flip-Flop
- N+1 Flip-Flop

II. Zero has a single representation in
- 1's Complement
- Sign-magnitude
- 2's Complement
- None of these

III. In case of binary subtraction(A-B) ,if A=B then value of End carry(E) is
- -1
- 1
- 0
- None of these

IV. The value of bias in IEEE754 double precision is
- 1023
- 0
- 1
- 127

V. The decimal roots of equation $x^2-(101)_b+(210)_b=0$ are 7 and 3.The value of b is
- 3
- 2
- 8
- 11

VI. How many check bits are needed to detect single bit error in 2000 bits data word?
- 20
- 12
- 11
- 10

VII. How many full adders require multiplying n-bits with m-bits in array multiplier?
- (n-1)*m
- n*(m-1)
- n*m
- None of these

VIII. Division algorithm in computer based on
- Compare ,Shift and subtract micro-operation
- shift and subtract micro-operation
- Compare ,Shift and add micro-operation
- shift and add micro-operation

SECTION-B (Attempt any two) **(3*2=6)**

I. How can we reduce carry propagation delay in binary adder? Justify your answer.

II. Suppose an 11-bit data word stored in memory is 00111101110.Using hamming algorithm, determine what check bits would be stored in memory with the data word?

III. Design an array multiplier that multiplies two 3-bit numbers. How many AND gates require to design it?

SECTION-C (Attempt any two) **(5*2=10)**

I. Describe block diagram of binary addition -subtraction algorithm for sign-magnitude representation .Drive an algorithm in flowchart form for adding and subtracting for two binary numbers when numbers are in sign-magnitude representation.

II. Show the step-by-step multiplication process of (-5)*(17) by using Booth's Multiplication.

III. Which architecture is based or stored program concept? What is major function of computer? List and briefly describe the possible states that define an instruction execution.

GLA University, Mathura

SET-C

I-Mid Term Examination, 2012-13

Course: - B.Tech. (CS) | **II-Year, III-Sem.**
Subject: - Discrete Mathematics (CSE-213) | **Uni. Roll No:-**
Time:- 90 Minutes | **Total Marks:-20**

Notes:-

1) *Answer all questions from Section A, **Any two** from Section B and **Any two** from Section C.*
2) *All parts of a question (a, b, etc.) should be answered at one place.*
3) *Answer should be brief and to-the-point and be supplemented with neat sketches.*
4) *Any missing or wrong data may be assumed suitably giving proper justification.*
5) *Figures on the right-hand side margin indicate full marks.*

SECTION-A

Objective Questions (Write the correct answer) : **(8*0.5 = 4)**

I. If two sets A and B have no common elements i.e. (A intersection B) has no element then such sets are known as ?
- Complement
- Disjoint
- Intersection
- Union

II. The domain D of the relation R is defined as the
- Set of all first elements of ordered pair which belongs to R
- Set of all elements of ordered pair which belongs to R
- Set of all last elements of ordered pair which belongs to R
- None of these

III. How many relations can be defined from set A containing m elements to set B containing n elements.
- $2^m * ^n$
- m*n
- $2^m * 2^n$
- None of the above

IV. The symmetric difference of A and B is
- $(A - B) \cup (B - A)$
- $(A \cup B) - (A \cap B)$
- $(A \cap B') \cup (A' \cap B)$
- All of the above

V. Determine the values of the constants A and B so that $a_n = An + B$ is a solution of the recurrence relation $a_n = 2a_{n-1} + n + 5$.
- 1 and -7
- -1 and -7
- -1 and 7
- 1 and 7

VI. The generating function for the sequence 1, 0, 1, 0, 1, 0, 1, 0, ...is
- $1/(1 - x^2)$
- $(1/(1 - x))^2$
- $1/(1+x)$
- $1/(1 + x^2)$

VII. Two dice are thrown simultaneously. The probability of getting a multiple of 3 as the sum is
- 1/6
- 1/2
- 1/3
- 5/6

VIII. If P (A) = 3/8, P (B) = 1/3 and P (A U B) = 1/4, then the value of P (A' U B') is
- 13/24
- 1/3
- 1/8
- 11/24

SECTION – B (Attempt any two) **(3*2=6)**

I. Prove that : $A - (B - C) = (A - B) \cup (A \cap C)$

II. Among the positive integers less than or equal to 100, let A be the set of even integers and let B be the set of integers divisible by 5. How many integers are even or divisible by 5?

III. A problem in mathematics is given to three students A, B, C, whose chances of solving it are 1/2, 1/3, 1/4 respectively. What is the probability that (i) the problem will be solved (ii) the problem will not be solved.

SECTION – C (Attempt any two) **(5*2=10)**

I. Let R be the relation on the set of ordered pairs of positive integers such that ((a , b) , (c , d)) $\in$ R if and only if ad = bc. Show that R is an equivalence relation.

II. Solve the recurrence relation $a_n = 3a_{n-1} + 4^{n-1}$ for $n \geq 0$ with $a_0 = 1$

III. Solve the recurrence relation by the method of generating functions $a_n - 9a_{n-1} + 20a_{n-2} = 2$ for $n \geq 2$ with initial conditions $a_0 = 1$, $a_1 = 2$.

GLA University, Mathura

SET - D

Mid I-Term Examination, 2012-13

Course: - B.Tech.(CS) II-Year, III-Sem.

Subject: - Data Structures & Algorithms (CSE211) **Uni. Roll No:-**

Time:- 90 Minutes **Total Marks:-20**

Notes:-

1) Answer all questions, Group A is **compulsory, Any Two** from Group B and **Any Two** from Group C.
2) All parts of a question (a, b, etc.) should be answered at one place.
3) Answer should be brief and to-the-point and be supplemented with neat sketches.
4) Any missing or wrong data may be assumed suitably giving proper justification.
5) Figures on the right-hand side margin indicate full marks.

GROUP-A **(0.5x8=4 Marks)**

Q1. What operation does the following code performs?

```
Node *node=head, *prev=0, *newnode;
while(node && node->data<newnode->data)
{ prev=node;
node=node->next;
}
```

➢ Find place to insert new node ➢ Assign node to prev ➢ Traverse the linked list ➢ None of these

Q2. What is the time complexity of the inserting n elements in a queue

➢ O(n^2) ➢ O(n) ➢ O(nlogn) ➢ O(1)

Q3. A solution to a 64 disks tower of hanoi problem requires how many disks to be moved?

➢ (64*2)-1 ➢ 64 ➢ (2^64-1) ➢ None of these

Q4. What is the Big-oh complexity of the following nested loop?

```
for(int i=0;i<n;i++)
{   for(int j=0;j<n;j++)
   {
    statements;
   }
}
```

➢ O(n^2) ➢ O(nlogn) ➢ O(n) ➢ O(1)

Q5. What would be output of the following code?

```
#include<stdio.h>
main()
{
char *ptr="abcd";
char ch =++*ptr++;
printf("\n%c",ch);
}
```

➢ c ➢ a ➢ b ➢ d

Q6. What does the term **j means?

➢ Pointer to pointer ➢ Value of j ➢ Pointer ➢ None of these

Q7. Which of the following data structure may give overflow error, even though the current number of element in it is less than its size?

➢ Priority queue ➢ Circular queue ➢ Simple queue ➢ None of these

Q8. What is the postfix form of the following prefix expression -A/B*C^DE

➢ A-BCDE^*/ ➢ ABC^ED*/- ➢ ABCDE^*/- ➢ A-BCDE^*/-

GROUP-B (2x3=6 Marks)

Q1. Determine what the following recursive C function computes. Write an iterative function to accomplish the same purpose.

```
func(n)
int n;
{ if(n==0)
return(0);
return(n + func(n-1));}
```

Q2. Translate following infix expression to their equivalent postfix expression showing the usage of stack for the conversion.

(X+Y-Z)/(H+K)*S

Q3. Write a program to insert a node after a given element in circular doubly linked list.

GROUP-C (2x5=10 Marks)

Q1. Write a program to reverse singly linked list.

Q2. Explain Tower of Hanoi problem for n=3.

Q3. If an array holding a queue is not considered circular, the text suggests that each remove operation must shift down every remaining element of a queue. An alternative method is to postpone shifting until rear equals the last index of the array. When that situation occurs and an attempt is made to insert an element into the queue, the entire queue is shifted is shifted down, so that the first element of the queue is in position 0 of the array. What are the advantages of this method over performing a shift at each remove operation? What are the disadvantages? Rewrite the routines remove and insert using this method.

BEST OF LUCK

D

GLA University
I Mid Term Exam
B.Tech., 3th semester (EC & CS)
Ethics & Values (AHE: 210)

Marks: 20 Time: 1.30 Hours

This paper consists of three passages. Carefully read the passages and answer the following multiple choice questions. Questions from **passage 1 carry 1 mark**, while questions from **passage 2 & 3 carry 1.5 marks**.

Passage-1

Morality can be comprehended simply in a manner we design and what we believe as right codes of conduct. Morality by defining what is right and good in a particular situation helps us to make the choices and perform the actions in this world. Morality defines what ought to be done to live the social life in a proper fashion. In moral context, the term value has a precise meaning and that is to evaluate the human conduct and practices. Value judgment to someone simply means diagnosing whether values (truth, respect, honesty, justice, etc...) are present in his/her character or not. Values are so closely associated with morality, that in absence of them we cannot meaningfully talk about morality. Generally, we use morality and ethics interchangeably and do not make any difference between these two terms. However, ethics differ from that of morality in a way it is moral philosophy in which we raise questions on morality by adopting a critical attitude to it. It is in this precise sense ethics is called a second order study. By ethical investigation we mean an argumentative critical examination based on various theories, reasons and individuals' point of views on rights and wrongs. Morality can be based on certain social and religious conventions but ethics follow reason in deciding what is right or wrong thing to do. Justice can be seen in terms of social welfare, individual freedom and virtues. It is a sort of an assessment of the morality of the society. To ask whether a society is just is to ask how it distributes the things we prize-income and wealth, duties and rights, powers and opportunities, offices and honors.

1) A just society works.
   a) for its welfare
   b) provides freedom to individuals
   c) cultivates virtue in individuals
   d) a, b, & c.
2) Morally proper conduct means:
   a) Right conduct
   b) Good conduct
   c) Necessary conduct
   d) a, b, & c.
3) Ethics is a second order study because
   a) It is a critical study.
   b) It examines the morality.
   c) It raises questions on morality.
   d) a, b & c.
   e)
4) Ethical inquiry involves:
   a) Critical attitude
   b) Rational attitude
   c) Individual point of views
   d) All above
5) Social and religious conventions can be used in defining what is:
   a) Morally appropriate thing to do.
   b) Ethically right thing to do.
   c) Both a & b.
   d) None of them.

Passage-2

Bentham, an English moral philosopher and legal reformer, founded the doctrine of utilitarianism. Its main idea is simply stated and intuitively appealing: The highest principle of morality is to maximize happiness, the overall balance of pleasure over pain. According to Bentham, the right thing to do is whatever will maximize utility. By "utility", he means whatever produces pleasure or happiness, and whatever prevents pain or suffering. For Bentham, pleasure is pleasure and pain is pain. The only basis for judging one experience better or worse than another is the intensity and duration of the pleasure or pain it produces. Bentham recognizes no qualitative distinction among pleasures. "The quantity of pleasure is equal", he writes, "push-pin is as good as poetry." The refusal to distinguish higher from lower pleasure is connected to Bentham's belief that all values can be measured and compared on a single scale. But some object to utilitarianism on precisely this point: they believe that some

pleasures really are "higher" than others. If some pleasures are worthy and others base, they say, why should society weigh all preferences equally, much less regard the sum of such preferences as the greatest good?

1) Bentham's utilitarian principle goes against our common human experience because:
   a) Bentham recognizes no qualitative distinction among pleasures.
   b) It maximizes utility.
   c) It does not care about the preferences of pleasures and weigh all preferences equally.
   d) a & c.
2) Intensity and duration are the bases to judge
   a) Pleasures and Pains
   b) Maximum utility
   c) One experience better or worse than another
   d) None of them.
3) The right thing to do according to utilitarianism
   a) Whatever will maximize happiness
   b) Whatever will maximize utility
   c) Whatever produces happiness or reduces suffering
   d) All above.
4) Bentham refuses to distinguish higher from lower pleasure because
   a) Pleasure is pleasure and pain is pain.
   b) All values can be measured and compared on a single scale.
   c) Some pleasures are higher than others.
   d) a & b.

Passage-3

The human faculties of perception, judgment, discriminative feeling, mental activity, and even moral preference, are exercised only in making a choice. He who does anything because it is the custom makes no choice. He gains no practice either in discerning or in desiring what is best. The mental and moral, like the muscular powers, are improved only by being used. He who lets the world, or his own portion of it, choose his plan of life for him, has no need of any other faculty than the ape-like one of imitation. He who chooses his plan for himself employs all his faculties. Mill concedes that following convention may lead a person to a satisfying life path and keep him out of harm's way. "But what will be his comparative worth as a human being?" he asks. "It really is of importance, not only what men do, but also what manner of men they are that do it. So actions and consequences are not all that matter after all. Character also counts.

For J. S.Mill, individuality matters less for the pleasure it brings than for the character it reflects. "One whose desires and impulses are not his own, has no character, no more than a steam engine has character. Mill put emphasis on liberty and higher order pleasures.

1) Rational choice making involves:
   a) Moral preference
   b) Judgment
   c) Discriminative feeling
   d) All above

2) When author says "no more than a steam engine has character" what he means:
   a) Steam engine has no character
   b) Stem engine has no desires and impulses
   c) Steam engine has character
   d) a & b
3) Author uses the phrase "the ape like imitation" for persons:
   a) Who follow custom
   b) Who choose his path of life for himself
   c) Who do not use mental and moral capacities
   d) All above
4) For J. S. Mill action and consequences are not all that matter. He put more emphasis on;
   a) Individuality & Character
   b) Higher Pleasure & Liberty
   c) a & b
   d) only b
5) He who follows customs;
   a) Makes moral choice
   b) Makes a cultural choice
   c) Makes no choice
   d) None of them
6) Conventions may guide us to the satisfying path of life but living life only by following conventions we lose;
   a) Meaning of human life
   b) Comparative worth as a human being
   c) Self belief
   d) b & c

A

GLA University, Mathura

Uni. Roll No:-

I-Mid Term Examination, 2012-13

Course: - B.Tech. II Year III Sem.

Subject: - MATHEMATICS - III (AHM – 201) Time:- 90 Minutes Total Marks:- 20

GROUP A (8 x 0.5 = 4)

Note : ALL questions are **COMPULSORY**. Choose and Write the **CORRECT** alternative on your answer sheet.

Q.1. The function $F(x) = \begin{cases} x + \pi \ ; & -\pi \le x \le 0 \\ -x + \pi \ ; & 0 \le x \le \pi \end{cases}$ is a / an

- Even function in $[-\pi, \pi]$
- Odd function in $[-\pi, \pi]$
- Neither even nor odd function in $[-\pi, \pi]$
- None of these

Q.2. If $f(x) = |\cos x|$ is expanded in a Fourier series in $(-\pi, \pi)$ then a_0 is equal to

- $4/\pi$
- $2/\pi$
- $\pi/4$
- None of these

Q.3. The graph of an odd function is

- Asymmetrical
- Symmetrical about x – axis
- Symmetrical about y – axis
- None of these

Q.4. A polynomial of degree n is called a complete polynomial if

- it contains all powers of the variable from 0 to n
- it does not contain all powers of the variable from 0 to n
- it contains powers of the variable from 0 to n – 1.
- None of these

Q.5. At a point of discontinuity of a function f (x), the value of f (x) given by Fourier series is

- Arithmetic mean of left and right limits
- Geometric mean of left and right limits
- Harmonic mean of left and right limits
- None of these

Q.6. If we want to remove the second term of a polynomial equation of degree n given by

$$a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \ldots\ldots\ldots + a_n = 0$$

then we divide the polynomial by x – h where h is given by

- $na_1 + ha_0 = 0$
- $na_0 + ha_1 = 0$
- $nha_0 + a_2 = 0$
- $nha_0 + a_1 = 0$

Q.7. The cubic equation whose roots are 6 times the roots of the equation

$$2x^3 + 5x^2 + x - 2 = 0 \text{ is}$$

- $y^3 + 18y^2 + 15y - 216 = 0$ where $y = 6x$
- $y^3 + 15y^2 + 18y - 216 = 0$ where $6x + y = 0$
- $y^3 + 15y^2 + 18y - 216 = 0$ where $6x - y = 0$
- none of these

Q.8. The period of function $f(x) = \cos x + (1/2)\cos 2x + (1/3)\cos 3x$ is

- π
- $2\pi/3$
- $(1/3)[2\pi + \pi + 2\pi/3]$
- 2π

GROUP B (2 x 3 = 6)

Note : Attempt **ANY TWO** questions.

Q.1. Solve by Cardon's method : $x^3 - 30x + 133 = 0$

Q.2. Find a Fourier series for the function

$$F(x) = \begin{cases} -1, & \text{for } -\pi < x < 0 \\ 0, & \text{for } x = 0 \\ 1, & \text{for } 0 < x < \pi \end{cases}$$

Hence deduce that $1 - (1/3) + (1/5) - (1/7) + \ldots\ldots\ldots = \pi/4$

Q.3. Solve the biquadratic equation: $x^4 - 12x - 5 = 0$ by Descartes' method.

GROUP C (2 x 5 = 10)

Note : Attempt **ANY TWO** questions.

Q.1. Obtain Fourier series for the function :

$$F(x) = \begin{cases} \pi x, & \text{if } 0 \le x \le 1 \\ \pi(2 - x), & \text{if } 1 \le x \le 2 \end{cases}$$

Q.2. Obtain the Fourier series to represent $F(x) = (1/4)(\pi - x)^2$ in the interval $[0, 2\pi]$.

Hence or otherwise evaluate : $\sum_{n=0}^{\infty} [1/(2n+1)^2]$

Q.3. Solve the following biquadratic equation by Ferrari's method :

$$x^4 - 3x^2 - 42x - 40 = 0$$